

Glycemic Index

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- We know that some carbohydrates are rapidly digested and absorbed, some are digested slowly while some are not digested at all.
- Thus, it is obvious that different carbohydrates will raise blood glucose levels to a different extent.
- The ability of carbohydrates to raise blood glucose is referred to as "Glycemic Index" (GI). GI is a quantitative assessment of foods based on postprandial blood glucose response (i.e. blood glucose level after a meal) expressed as a 'percentage of the response to an equivalent carbohydrate portion of a reference food'.
- The reference food is white bread with a GI set at 100.

- Knowing about GI of foods helps us in prescribing therapeutic diets like diabetic and energy-controlled diets.
- We realize that the dietary GI provides an indication of the rate at which, carbohydrate foods are digested. It allows ranking of foods from those which give rise to the highest blood glucose and insulin responses (high glycemic food) to those associated with the lowest blood glucose and insulin responses (low GI foods).
- High GI foods lead to a rapid rise in blood glucose and insulin levels. Hyper-insulinemia in turn may down regulate insulin receptors and therefore reduce insulin efficiency resulting in insulin resistance.

- On the contrary, low GI foods result in reduced peak insulin concentrations and reduced insulin demand thus decreasing the risk of type II diabetes.
- The health benefits include reduced insulin demand, improved blood glucose control and reduced blood lipid levels.
- All these factors play an important role in the prevention and management of chronic diseases including diabetes, coronary heart disease and certain cancers.

Nutritional requirement and health aspects

- In our daily diet almost 60-70% of energy is contributed by carbohydrates.
- Majority of them are comprised of starch (from cereals, millets, pulses and root vegetables), while a small amount is also contributed by sucrose (sugar), glucose and fructose (fruits), as well as lactose (milk and milk products).

Deficiencies

- Hypoglycemia –Leads to weakness due to insufficient carbohydrates and may cause unconsciousness
- Body starts using protein as fuel for energy. So body loses muscle mass. Such a Protein usage also leads to stress on kidneys.
- Insufficient fiber content leads to constipation.